How Competitive is Modern Sumo Wrestling?

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1 Introduction

In this paper, we assess the competitive balance of professional sumo wrestling over a period of nearly 25 years. Sumo wrestling has many features that make it unique compared to other sports. It is an individual sport where a large group of athletes face off against their competitors many times per year at a variety of locations. Specifically, there are six tournaments per year, and in the top two professional divisions (which we examine in this paper), each wrestler has 15 bouts per tournament. Sumo wrestling has a total of six divisions, with the top two divisions (Makuuchi and Juryo) being considered the serious professional levels. After each tournament, wrestlers are promoted or demoted to different ranks (within or across divisions) based on their performance, with additional ranks being available to the highest performing athletes. Further, professional sumo wrestling has no weight or height categories (unlike many other combat sports), which can lead to significant mismatches in size and strength between competitors. Another unique feature of sumo wrestling is the short "game time" in each bout, which may only last a few seconds. A fight is considered over when one wrestler exits the ring or touches the ground with any part of their body other than the soles of their feet. Since wrestlers can feint or dodge their opponents and due to the very short duration of the fight, it may seem that randomness is

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a significant factor in determining the winner of a sumo bout.

Since each wrestler fights 90 times per year against a wide variety of opponents, there are many games and matchups to analyze. Fortunately, a public database posts the results of every professional sumo bout, along with the ranks, heights, weights, etc. of each wrestler in advance of every tournament. This gives us a rich set of data to analyze over a long period of time.

2 Literature Review

2.1

2.2

2.3

3 Methods

3.1 Data Preparation

3.2 Modeling

4 Results and Analysis

4.1 Summary Analysis

4.2 Correlation Analysis

4.3 Models

4.4 Model Predictions

All results in this section use the LightGBM model with the highest R^2 from the table above, unless otherwise specified.

4.4.1 Sentiment Contribution

5 Discussion

6 Conclusion

7 Appendix